HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and executed thermal analysis for air-cooled chiller components, including heat exchanger sizing and refrigerant circuiting, contributing to **EN 14511 and EN 14825** compliance documentation.
- Processed and plotted laboratory test data for chiller performance, quantifying deviations against modelled results and reporting findings to the product team to refine **performance ratings**.
- Built and maintained spreadsheet-based analysis tools for parametric studies and curve fitting of compressor and heat exchanger data, extending the capabilities of production **rating programs**.
- Supported engineering efforts by preparing performance data for European products, aligning with **Ecodesign / ErP compliance** and Eurovent certification requirements.
- Analyzed the impact of control strategies, such as compressor staging and fan speed control, on **seasonal efficiency** and rated performance for new chiller designs.
- Collaborated with global Modeling & Simulation teams to provide feedback on production models and rating engines, highlighting data gaps and requesting new capabilities based on European market needs.
- Containerized analysis tools and supporting scripts using **Docker** for consistent deployment across development and testing environments on **AWS EC2**.

SKILLS

Engineering Concepts – Thermodynamics, Heat Transfer, Fluid Principles, Vapor Compression Cycle, Refrigeration Systems, Air-Cooled Chillers

Performance & Compliance – EN 14511, EN 14825, Ecodesign / ErP, Eurovent Certification, CE Marking, PED, EU F-Gas Regulation

Analysis & Tools – Spreadsheet Analysis, Data Reduction, Curve Fitting, Parametric Studies, Regression Analysis, Compressor Performance Maps, Heat Exchanger Performance

Programming & Scripting – Python, SQL, R, MATLAB

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker

Data Handling – Laboratory Test Data Analysis, Large Data Set Processing, Data Visualization

PROJECTS

Chiller Performance Rating Tool Development

- Engineered a Python-based tool to automate the processing of raw laboratory test data for air-cooled chillers, reducing manual effort by approximately **40%**.
- Implemented curve fitting algorithms in Python to characterize compressor performance maps and heat exchanger data, enabling more accurate prediction of operational envelopes.
- Developed interactive visualizations using Matplotlib and Seaborn to compare modelled performance against test results, aiding in the identification of discrepancies for further investigation.

European Air-Cooled Chiller Efficiency Analysis

- Performed detailed thermal analysis on various air-cooled chiller configurations to support compliance with **EN 14511 and EN 14825** standards for the European market.
- Built a parametric study framework in MATLAB to evaluate the impact of component selection and operating conditions on SEER and SEPR ratings, across a range of ambient temperatures.
- Generated comparative capacity and efficiency data for different refrigerant transitions, including **R-454B and R-1234ze(E)**, to inform product strategy.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)